tcs | **TCS AI HACKATHON**
THE NEXT WAVE OF INTELLIGENCE

THEME 2
AI for Business

DOMAIN
Financial Crime

CHALLENGE 07
AI Financial Crime Network Investigator

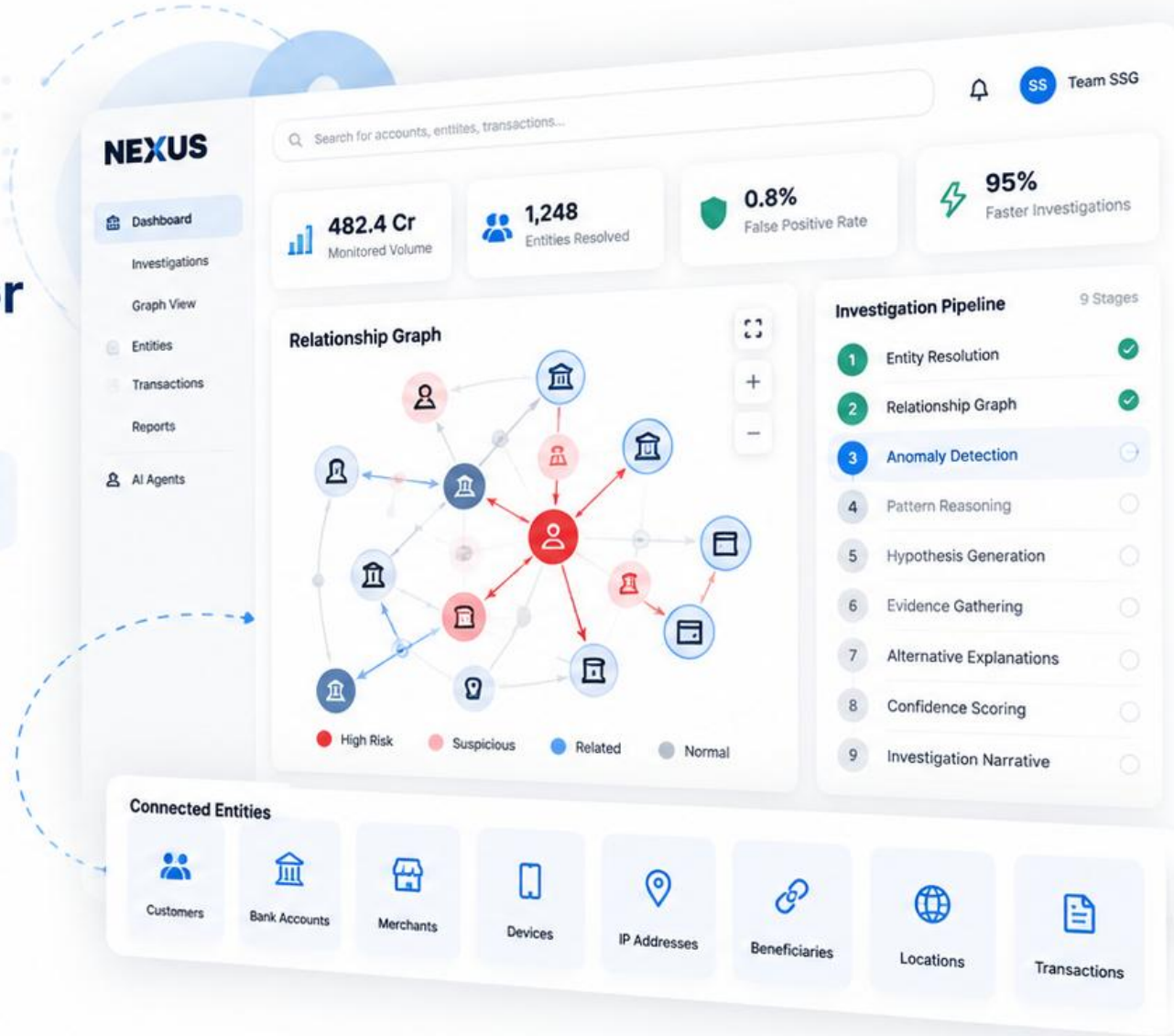
NEXUS

AI Financial Crime Network Investigator

Uncovering Hidden Connections. Enabling Smarter Investigations.

- Graph AI
- Fraud Investigation
- Explainable AI
- Agentic AI

An AI-driven platform that discovers, analyzes and explains suspicious financial networks across customers, accounts, merchants, devices, IP addresses and transactions.



TEAM SSG
TCS AI HACKATHON

Our Vision

To transform fragmented financial signals into clear, explainable insights for faster and more accurate investigations.

PROBLEM STATEMENT

Financial Crimes Are Becoming More Complex

Suspicious signals are distributed across multiple sources, making it difficult to detect hidden relationships.

Fragmented Information

Transaction data, accounts, merchants, devices, IP addresses and beneficiaries are stored in silos.

Hidden Relationships

Criminal networks use multiple accounts, devices, and intermediaries to disguise fund flows.

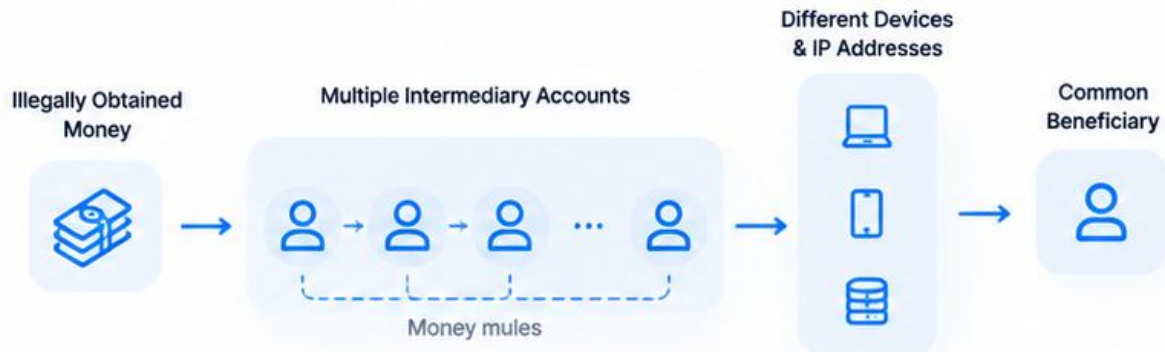
Manual Investigation is Slow

Reviewing each signal independently makes it time-consuming and error-prone.

High False Positives

Traditional systems generate many alerts, creating manual effort for investigators.

How Financial Crime Networks Operate



The Challenge

Build an AI investigator that can discover and reason about suspicious relationships across customers, bank accounts, merchants, devices, IP addresses, beneficiaries, locations and transactions — and explain why the network is suspicious.

OUR SOLUTION

NEXUS – AI Financial Crime Network Investigator

From scattered alerts to comprehensive network intelligence.

Discover Connected Networks
Links customers, accounts, merchants, devices, IP addresses and transactions to uncover hidden relationships.

01

Explain Why It's Suspicious
Identifies unusual patterns and provides clear reasons with supporting evidence and risk score.

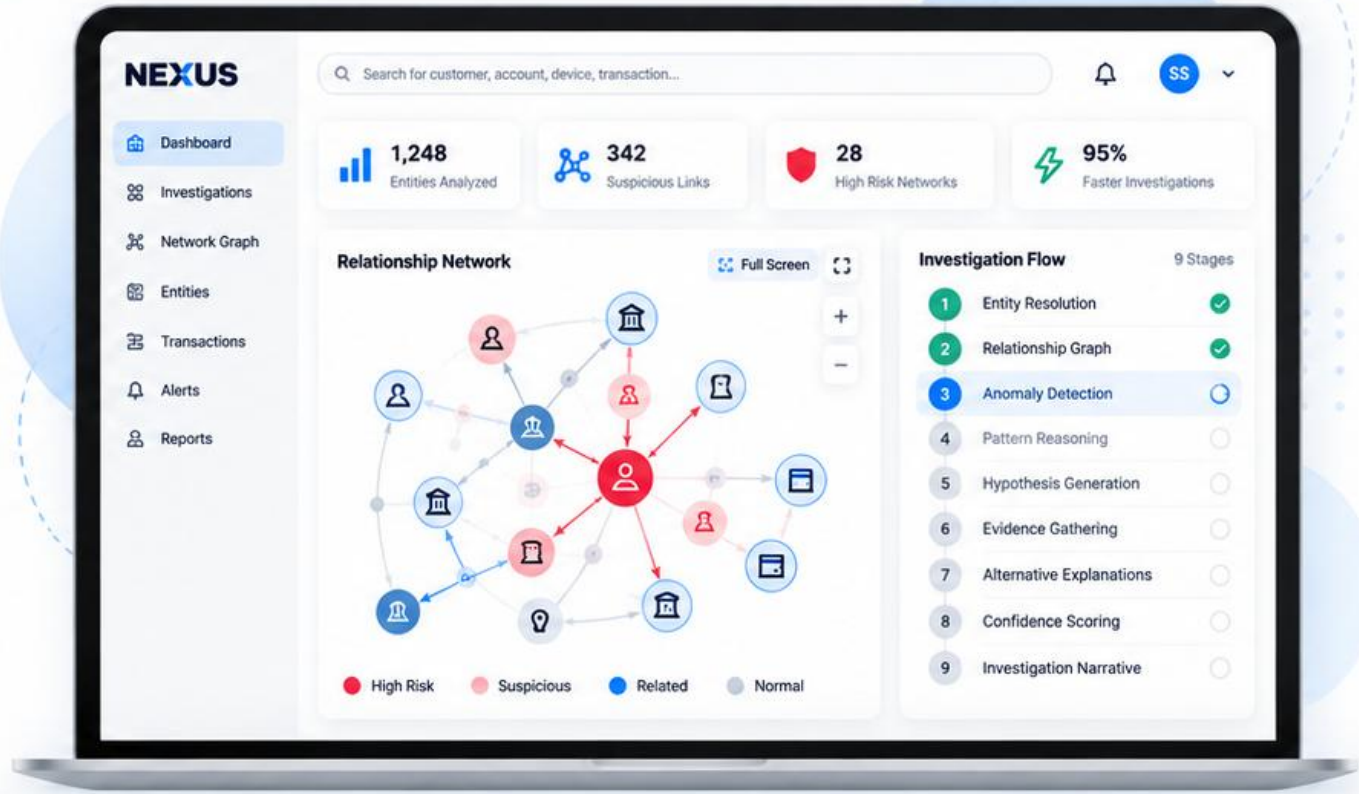
02

Generate Investigation Insights
Creates hypotheses, considers alternative explanations and highlights what to investigate next.

03

Enable Faster, More Informed Decisions
Brings all relevant information into a single view to help investigators act with confidence.

04



End-to-End Investigation Workspace
From network discovery to investigation narrative.

Explainable & Transparent
Clear reason, evidence and confidence for every conclusion.

Human-in-the-Loop
Supports investigators with insights, not autonomous decisions.

Measurable Business Impact
Reduced investigation time, lower false positives, better risk visibility.

TECHNICAL ARCHITECTURE

End-to-End System Architecture

A modular, scalable architecture to discover, analyze and explain financial crime networks.

1 Data Ingestion Layer

Multiple enterprise data sources

- **Core Banking Transactions** (ISO 8583 / ISO 20022)
- **Merchant / POS Data** (Transactions, Category)
- **Device & Network Data** (IMEI, IP, Location)
- **Customer & KYC Data** (Accounts, Beneficiaries)
- **External Sources** (Sanctions, Complaints, etc.)

2 API & Orchestration Layer

Handles data processing and workflow orchestration

Backend Server
(Node.js + Express)
- API Endpoints
- Authentication & Access Control
- Request Routing

Processing & Orchestration
- Data Normalization
- Workflow Management
- Audit Logging

Database Layer
(MySQL / PostgreSQL)
- Transaction Data
- Entities & Relationships
- Investigation Results

3 AI Analysis Layer

AI models and domain agents for investigation

AI Service
(Python + FastAPI)
- Data Processing
- LLM Integration
- Analysis & Reasoning
- Guardrails & PII Redaction

Specialized AI Agents

- Entity Resolution
- Graph Analysis
- Anomaly Detection
- Pattern Reasoning
- Hypothesis Generation
- Evidence Discovery

4 Investigation Engine

9-stage forensic pipeline for network analysis

1. **Entity Resolution** — Link related entities
2. **Relationship Graph** — Build network connections
3. **Anomaly Detection** — Detect unusual patterns
4. **Pattern Reasoning** — Identify fraud typologies
5. **Hypothesis Generation** — Generate possible scenarios
6. **Evidence Gathering** — Collect supporting evidence
7. **Alternative Explanations** — Consider legitimate scenarios
8. **Confidence Scoring** — Calculate risk score
9. **Investigation Narrative** — Generate final report

5 Application Layer

Investigation workspace for analysts

- **Dashboard** — Key metrics and investigation overview
- **Network Graph** — Interactive relationship graph (Cytoscape.js)
- **Investigation Workspace** — 9-stage analysis view
- **AI Agent Console** — Interact with specialized agents
- **Report Generation** — Export investigation report (FIU-IND format)

Supporting Services

- **Authentication** & Authorization
- **LLM Services** (Gemini / Groq / DeepSeek)
- **Data Storage** (MySQL / PostgreSQL)
- **File Storage** (Evidence & Reports)
- **Monitoring & Logging** (System Performance)

Feedback & Continuous Improvement

BUSINESS VALUE

From Reactive Alerts to Proactive Intelligence

Delivers measurable impact for banks and financial institutions through faster, accurate and explainable investigations.

95%
Reduction in Investigation Time

>95% → 0.8%
Lower False Positive Rate

99%+
Entity Resolution Accuracy

₹X Cr
Potential Fraud Loss Prevention

Key Business Benefits

Faster Investigations
Automates network discovery and analysis, reducing manual effort.

Lower Operational Costs
Minimizes false positives, reducing compliance workload.

Better Risk Visibility
Identifies complex cross-entity networks and hidden risks.

Explainable & Auditable
Clear evidence trail and reasoning for every conclusion.

Regulatory Compliance Support
Helps generate structured reports for regulatory filings.

Traditional vs NEXUS

Traditional Approach	With NEXUS
✗ Isolated transaction alerts	✓ Network-wide analysis
✗ High false positives	✓ Significantly lower false positives
✗ Manual investigation	✓ AI-assisted investigation
✗ 2-3 days per case	✓ Results in minutes
✗ Limited explainability	✓ Clear reasoning & evidence
✗ High operational cost	✓ More efficient and scalable



Enabling Safer, More Transparent and Trustworthy Financial Systems

END USERS

Designed for the People Who Protect the Financial System

NEXUS supports multiple stakeholders across the financial crime investigation ecosystem.



Fraud Investigation Team
Use NEXUS to detect and analyze suspicious networks and investigate cases faster.



Compliance Officers
Monitor high-risk activities, ensure regulatory compliance and review AI-generated insights.



Risk Management Teams
Gain enterprise-level visibility into emerging risks and trend patterns.



Regulatory Authorities
Review etructured reports, evidence and investigation narratives for regulatory action.

A Unified Investigation Workspace

Role-based access with tailored views and tools for each user type.



How End Users Benefit

Access relevant investigation views → Explore network insights → Review AI explanations and evidence → Take informed action

TECHNOLOGY STACK

Modern Tools for Real-World Impact

A powerful combination of modern technologies and AI tools to build an intelligent, scalable and reliable solution.



AI Tools & Models

Used for development, AI capabilities and UI/UX design

Antigravity
AI development environment

Gemini 3.8 Flash
LLM for analysis and reasoning

GitHub Copilot Pro
AI pair programming and code assistance

Claude Sonnet 4.5
Advanced reasoning and analysis

Banana AI
UI/UX design and prototyping

Stitch
UI design to code generation

Application Tech Stack

Technologies used to build and deploy the solution

Frontend

React
Frontend Framework

Tailwind CSS
UI Styling

Zustand
State Management

Backend

Node.js
Runtime Environment

Express.js
Web Framework

Database

PostgreSQL
Relational Database

AI & Orchestration

LangGraph
Agent orchestration and workflow

Guardrails
Security, safety and PII protection

FastAPI
AI service backend

Deployment & Others

Render
Cloud Deployment

GitHub
Version Control

Docker
Containerization

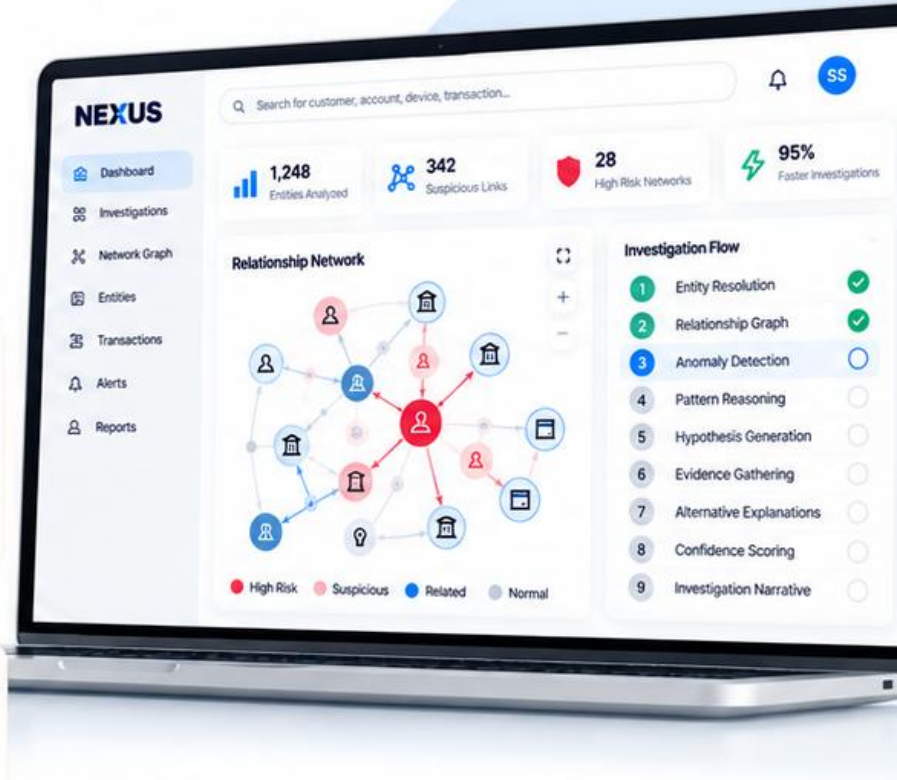
Built for Scalability, Security and Real Impact

- Modern & Scalable Architecture
- Secure & Responsible AI Usage
- Rapid Development with AI Tools
- Production Ready & Future Ready

CONCLUSION

Towards a Safer and More Transparent Financial Ecosystem

NEXUS transforms fragmented financial data into connected intelligence, empowering investigators to detect, understand and act on financial crime networks.



KEY TAKEAWAYS

Network Intelligence
Discovers hidden relationships beyond individual alerts.

Explainable Insights
Provides clear reasons with supporting evidence and alternative explanations.

Investigator Empowerment
Helps investigators focus on high-risk cases and make faster, confident decisions.

Real Business Impact
Reduces fraud losses, improves compliance efficiency and strengthens financial trust.

OVERALL OUTCOME

95%
Reduction in Investigation Time

0.8%
False Positive Rate

99%+
Entity Resolution Accuracy

Significant
Fraud Loss Prevention

Our Vision

To build a future where financial institutions can proactively prevent crime, protect legitimate customers and create a safer, more transparent financial ecosystem through responsible and explainable AI.

"From Hidden Networks to Safer Tomorrow."

NEXUS – AI Financial Crime Network Investigator